* Gray levels $= 2^K$ = bits to store on imge $= M*N*K$

* $K = \log_2(L)$

* Dynamic range = max : min = dark : bright = 10 : 90

* Spatial resolution = $W*H = 4*4$

* Intensity resolution = K.

* $D_e = \sqrt{(x-s)^2 + (y-t)^2}$ (actual)

* $D_4 = |x-s| + |y-t|$ ($\leftrightarrow \updownarrow$ maze)

* $D_8 = \max(|x-s|, |y-t|)$ ($\leftrightarrow \updownarrow \swarrow$ pathFinding, recognition)

* Translation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* Rotation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* Scaling:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* Vertical Flip:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* Horizontal Flip:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

* Skewing x:

$$\begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{aligned} x' &= x + sh_x \cdot y \\ y' &= y. \end{aligned}$$

* Skewing y:

$$\begin{bmatrix} 1 & 0 \\ sh_y & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{aligned} x' &= x \\ y' &= sh_y x + y. \end{aligned}$$

⇒ Histogram:

gray level, Frequency, $P_r = \frac{\text{count}}{M*N}$, CDF $= P_r Z$, equalized gray = CDF $*(L-1)$

⇒ Contrast stretching:

$$\alpha = \frac{s_1 - 0}{r_1 - 0}, \quad \beta = \frac{s_2 - s_1}{r_2 - r_1}, \quad \gamma = \frac{L-1-s_2}{L-1-r_2}$$

$$S = \begin{cases} \alpha * r & 0 \leq r < r_1 \\ \beta(r - r_1) + s_1 & r_1 \leq r \leq r_2 \\ \gamma(r - r_2) + s_2 & r_2 \leq r < L-1 \end{cases}$$

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⇒Image negative: enhances grey detail in dark regions

$$S = (L-1) - r.$$

⇒Logarithmic transformation: enhances details in dark, compresses light areas.

$$S = c * \log(1+r) \qquad c = \frac{L-1}{\log L}$$

⇒Gamma correction.

$$S = c * r^{\gamma}, \quad c = \frac{255}{255^{\gamma}}$$

$\gamma < 1$ to brighten (150, less)

$\gamma > 1$ to darken.

⇒Bit plane slicing: represent with bits change to binary and split to MSB, middle, LSB.

⇒3*3 Gaussian kernel:

$\frac{1}{16} *$

1	2	1
2	4	2
1	2	1

⇒3*3 Averaging kernel

$\frac{1}{9} *$

1	1	1
1	1	1
1	1	1

⇒5*5 circular kernel:

$\frac{1}{21} *$

0	1	1	1	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
0	1	1	1	0

⇒5*5 pyramidal Kernel:

$\frac{1}{81} *$

1	2	3	2	1
2	4	6	4	2
3	6	9	6	3
2	4	6	4	2
1	2	3	2	1

⇒Cone Kernel 5*5

$\frac{1}{25} *$

0	0	1	0	0
0	2	2	2	0
1	2	5	2	1
0	2	2	2	0
0	0	1	0	0

⇒Sobel operators:

$$mag = \sqrt{G_x^2 + G_y^2}$$

-1	-2	1
0	0	0
1	2	1

G_y

-1	0	1
-2	0	2
-1	0	1

G_x

2	1	0
1	0	-1
0	-1	-2

G_D

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⇒ Laplacian operator:

0	-1	0
-1	4	-1
0	-1	0

⇒ Metric quantity of How much an image has been compressed:

$$C = \frac{b}{b'} = \frac{\text{uncompressed size}}{\text{compressed size}}$$

⇒ Measuring compression Efficiency: $R = 1 - \frac{1}{C}$

⇒ Average code Length: $L_{avg} = \sum L(r_k) P(r_k)$

⇒ Entropy Equation: $H = -\sum P(r_k) \log_2 (P(r_k))$

⇒ Root Mean Square Error(RMSE): $\sqrt{\frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [f(i,j) - f'(i,j)]^2}$

⇒ Signal-to-Noise Ratio: $SNR = 10 \log_{10} \frac{\sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [f(i,j)]^2}{\sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [f(i,j) - f'(i,j)]^2}$

⇒ Erosion: $A \ominus B = [z | (B)_z \subseteq A]$ at pixel (x,y) = $\begin{cases} 1 & \text{if } B \text{ fits } A \\ 0 & \text{otherwise} \end{cases}$

⇒ Dilation: $A \oplus B = [z | (B)_z \cap A \neq \emptyset]$ at pixel (x,y) = $\begin{cases} 1 & \text{if } B \text{ Hits } A \\ 0 & \text{otherwise.} \end{cases}$

⇒ Compound operations:

Closing ⇒ $A \cdot B = (A \oplus B) \ominus B$

opening ⇒ $A \circ B = (A \ominus B) \oplus B$

⇒ Boundary Extraction:

$\beta(A) = A - (A \ominus B)$.

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$$\Rightarrow output_{norm} = \frac{output - output_{min}}{output_{max} - output_{min}} * 255$$

Note that:

O/P_{max} = largest pixel value calculated in solution

⇒ Image segmentation.

$$T_1 = \frac{1}{2}(\mu_1 + \mu_2)$$

note: calc. T_0, then T_1 & T_2 with the same rule. Compared to the previous T and see the differen $T_2 - T_1 = 0$, Repeat it until it reaches the least possible value.

$$\Rightarrow PSNR = 10 * \log_{10}\left(\frac{(255)^2}{MSE}\right)$$

⇒ Laplacian of Gaussian (LoG) 5*5 mask.

0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

⇒ 4-connected set

0	1	0
1	1	1
0	1	0

⇒ 8-connected set

1	1	1
1	1	1
1	1	1

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